

Feature Store — adoption review

ML PLATFORM / DECISION REVIEW

A feature store is a decision about ownership,
not about tooling.

PREPARED BY	DATE	AUDIENCE
PRESENTER · TEAM	2026.08	ML PLATFORM & DATA ENG

The problem a feature store solves is duplication, not storage

WHAT IT SOLVES

The same feature, written twice

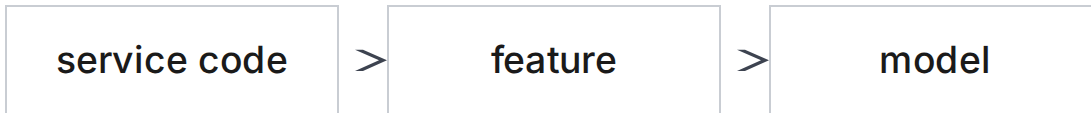
A feature computed one way for training and another way for serving will drift apart.

The failure is silent: the model does not error, it just gets quietly worse.

TRAINING PATH



SERVING PATH



One definition, two implementations — and nothing compares them.

A feature store is three services, and you may only need one

WHAT IT IS

01

Registry

One place where a feature is defined, owned and discoverable.

YOU NEED IT WHEN

You need it as soon as two teams use the same feature.

02

Offline store

Point-in-time correct history, used to build training sets.

YOU NEED IT WHEN

You need it if backfills keep leaking future data.

03

Online store

Low-latency reads of the same values at serving time.

YOU NEED IT WHEN

You need it only if models serve online requests.

Adoption depends on reuse and on whether serving is online

WHEN IT IS WORTH IT

Two conditions, and both must hold

Reuse alone is a registry problem. Online serving alone is a caching problem.

Only together do they justify running all three services.

↑ ONLINE SERVING NEED

Cache, not a platform

One model serving online does not need a store.

Adopt all three

This is the case the thing was built for.

Do not adopt

A table and a scheduled job are enough.

Adopt the registry only

Shared definitions, without an online store.

FEATURE REUSE ACROSS TEAMS →

The cost is a new critical dependency in the serving path

WHAT IT COSTS

01 A latency budget you now own

Every online read sits inside the request path.

02 A new page in the on-call rotation

The store failing means every model failing.

03 Backfill correctness

Point-in-time joins are easy to get subtly wrong.

04 Migration of what already exists

Existing features must move, or you run both.

What we need to decide before building or buying

DISCUSSION

01 How many teams share a feature definition today?

02 Do any of our models serve online requests?

03 Who would carry the page when the store is down?

Thank you

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